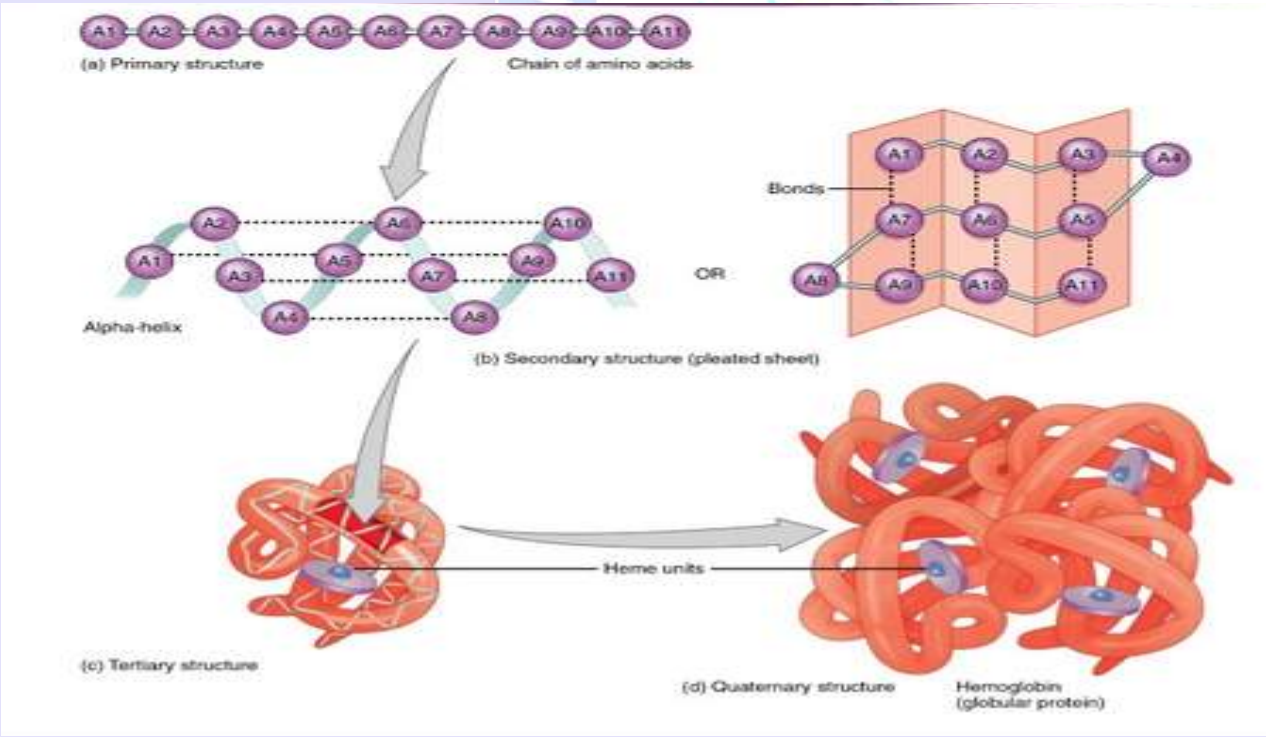
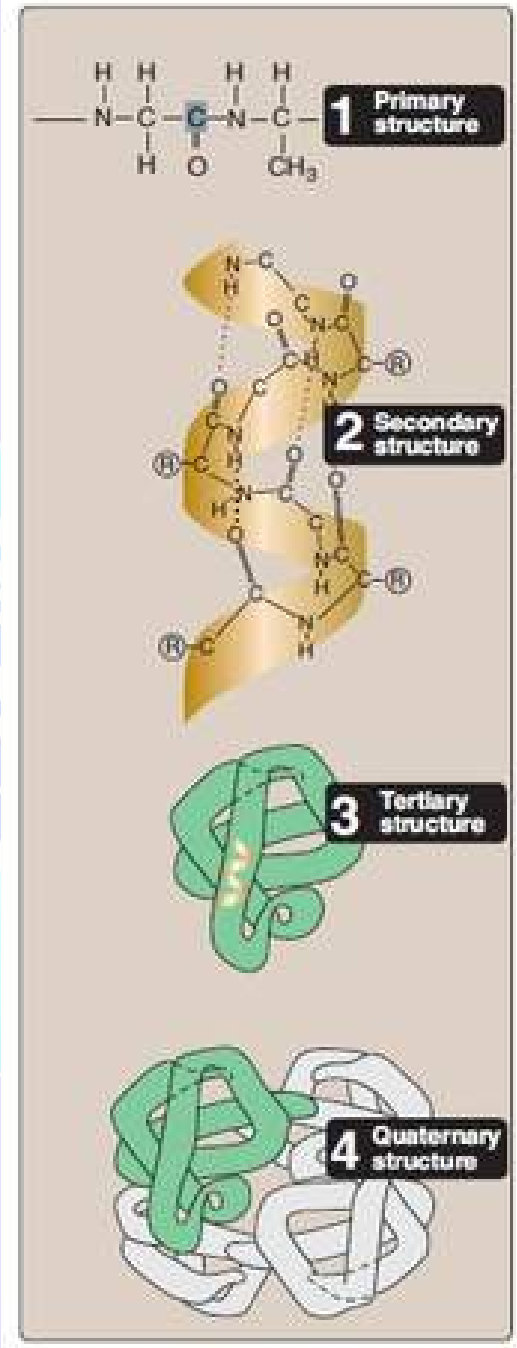
PROTEINS STRUCTURE



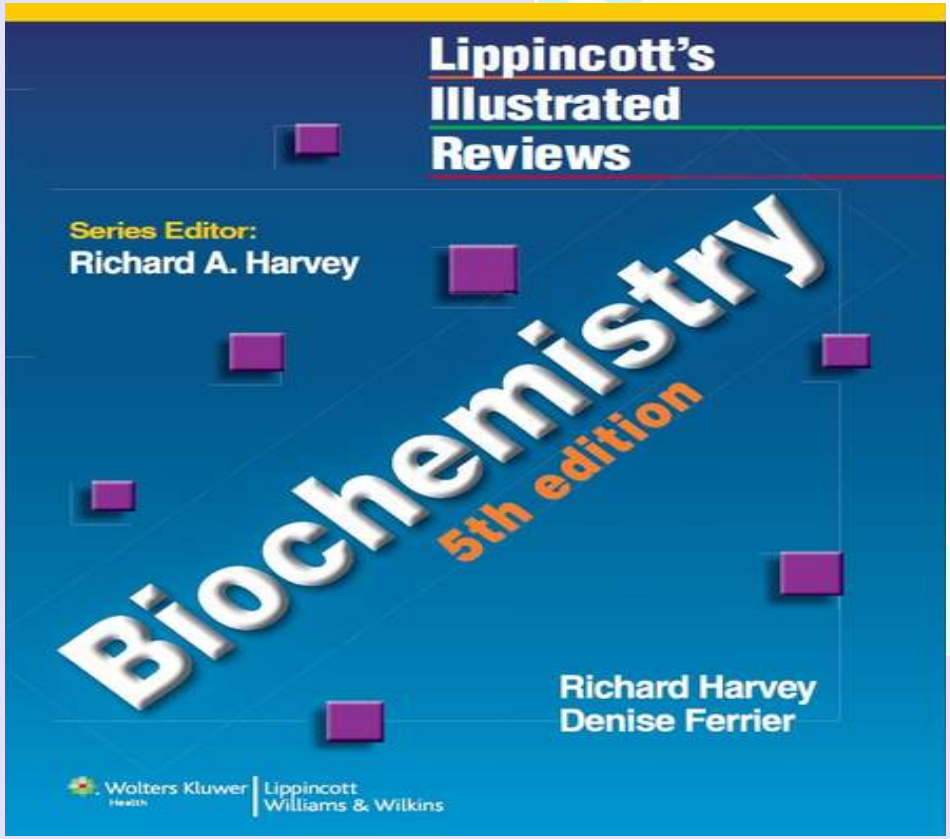


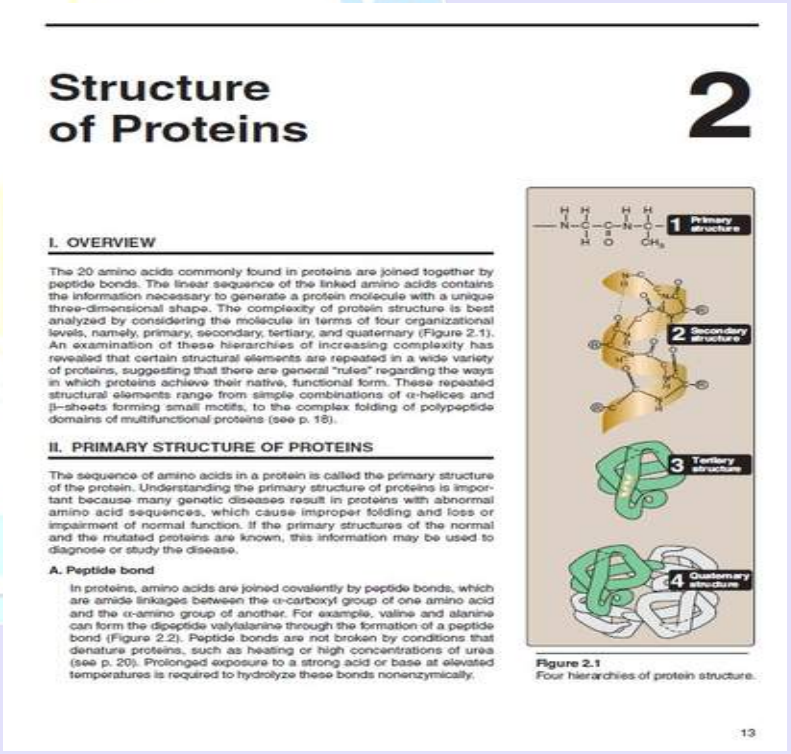
Learning Objectives

- At the end of this session the students should be able to:
  - Discuss structural level(s) of proteins
  - Define different bonding involve in maintaining structure(s) of proteins

Reference

Chapter 2 – Structures of Proteins. In: Lippincott's Illustrated Review Biochemistry 6th Edition.





Structure of Proteins 2

I. OVERVIEW

The 20 amino acids commonly found in proteins are joined together by peptide bonds. The linear sequence of the linked amino acids contains the information necessary to generate a protein molecule with a unique three-dimensional shape. The complexity of protein structure is best analyzed by considering the molecule in terms of four organizational levels, namely, primary, secondary, tertiary, and quaternary (Figure 2.1). An examination of these hierarchies of increasing complexity has revealed that certain structural elements are repeated in a wide variety of proteins, suggesting that there are general "rules" regarding the ways in which proteins achieve their native, functional form. These repeated structural elements range from simple combinations of α-helices and β-sheets forming small motifs, to the complex folding of polypeptide domains of multifunctional proteins (see p. 18).

II. PRIMARY STRUCTURE OF PROTEINS

The sequence of amino acids in a protein is called the primary structure of the protein. Understanding the primary structure of proteins is important because many genetic diseases result in proteins with abnormal amino acid sequences, which cause improper folding and loss or impairment of normal function. If the primary structures of the normal and the mutated proteins are known, this information may be used to diagnose or study the disease.

A. Peptide bond

In proteins, amino acids are joined covalently by peptide bonds, which are amide linkages between the α-carboxyl group of one amino acid and the α-amino group of another. For example, valine and alanine can form the dipeptide valylalanine through the formation of a peptide bond (Figure 2.2). Peptide bonds are not broken by conditions that denature proteins, such as heating or high concentrations of urea (see p. 20). Prolonged exposure to a strong acid or base at elevated temperatures is required to hydrolyze these bonds nonenzymically.

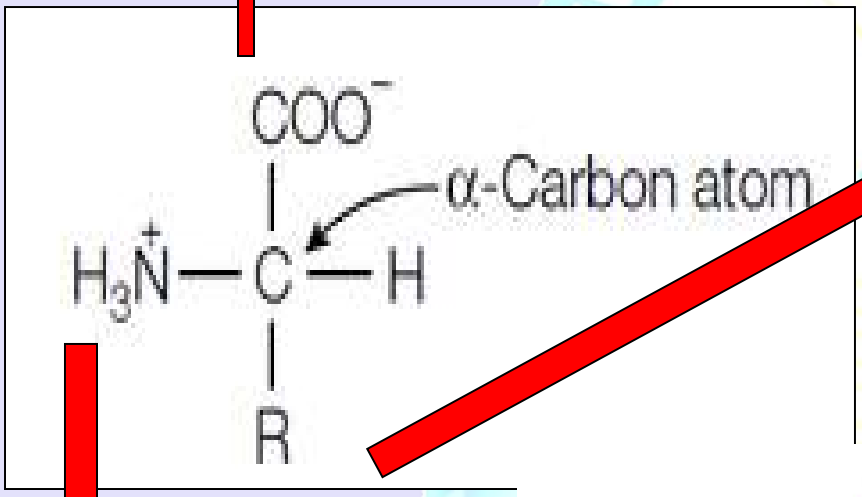
Figure 2.1
Four hierarchies of protein structure.

13

What are PROTEINS and why they are important to our body?

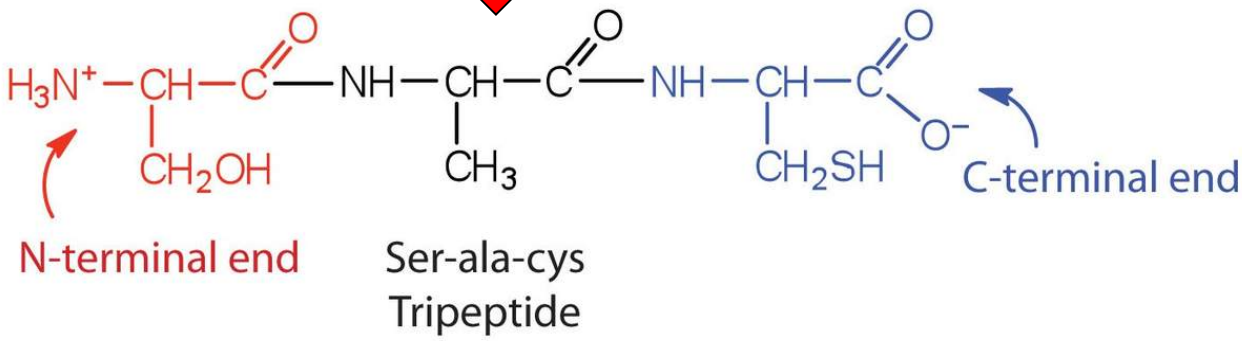
Amino acid: Basic unit of protein

Carboxylic acid group



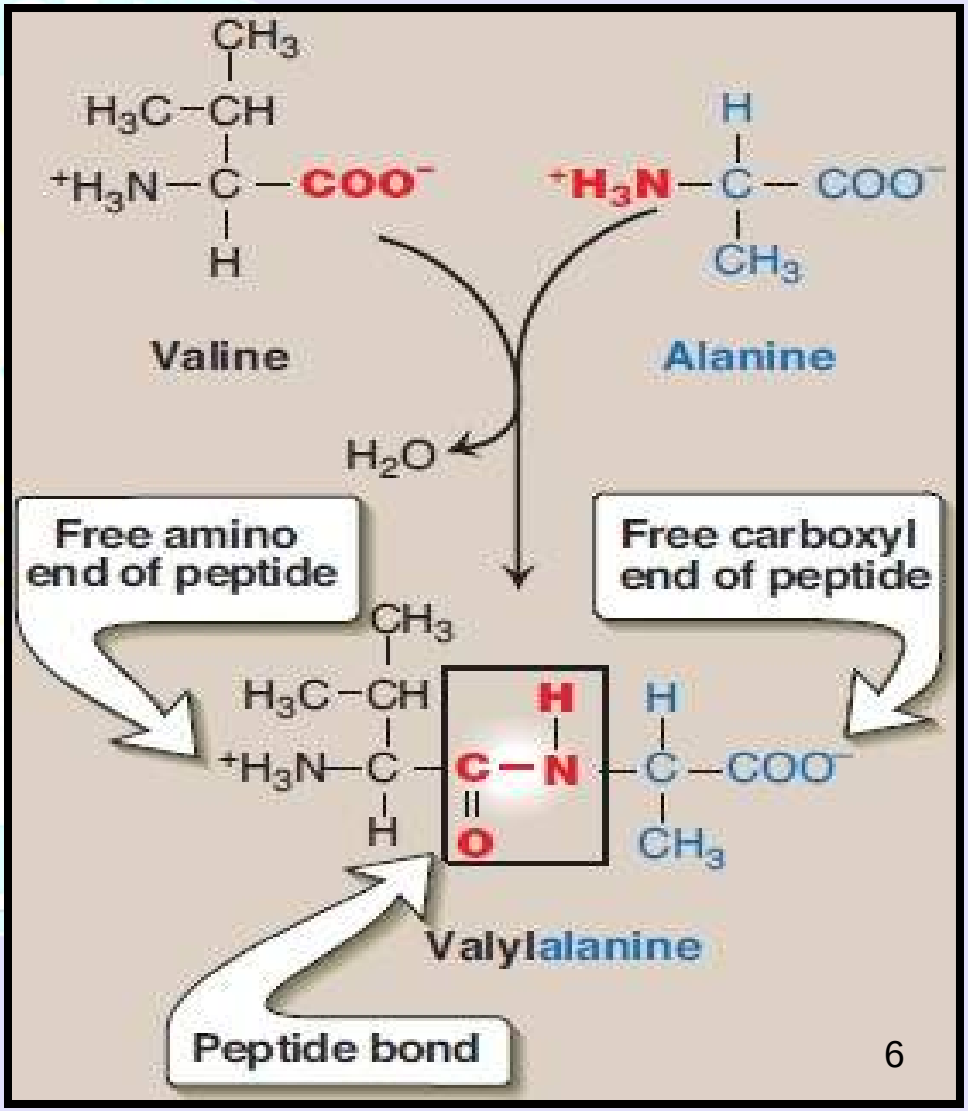
Amino group

Different side chains, R, determin the properties of 20 amino acids.



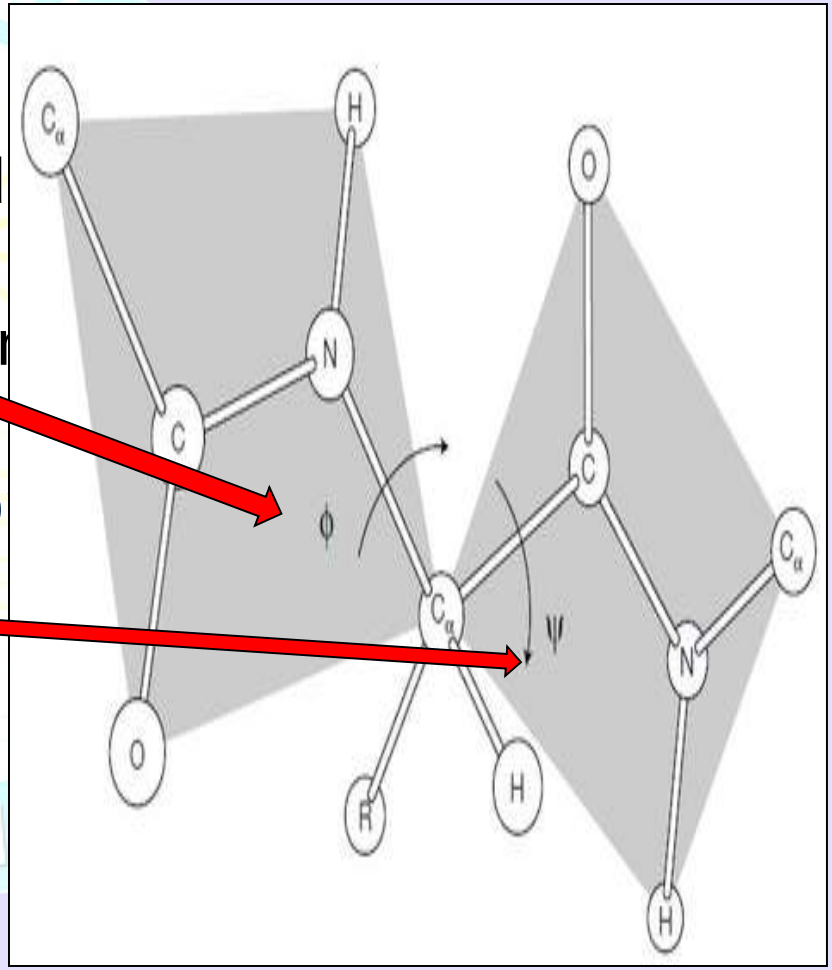
Protein & Peptide Bond

- Is a **covalent** bond
- are amide linkages between the α-carboxyl group of one amino acid and the α-amino group of another
- Peptide bonds are not broken by conditions that denature proteins



Rotational Properties of Peptide Bonds

- Rotation around the peptide bond is not permitted. It is rigid
- Rotation around bonds connected to the alpha carbon is permitted
- ϕ (phi): angle around the α-carbon —amide nitrogen bond
- ψ (psi): angle around the α-carbon —carbonyl carbon bond
- In a fully extended polypeptide, both ψ and ϕ are 180°
- Allows the polypeptide chain to assume a variety of possible configurations.

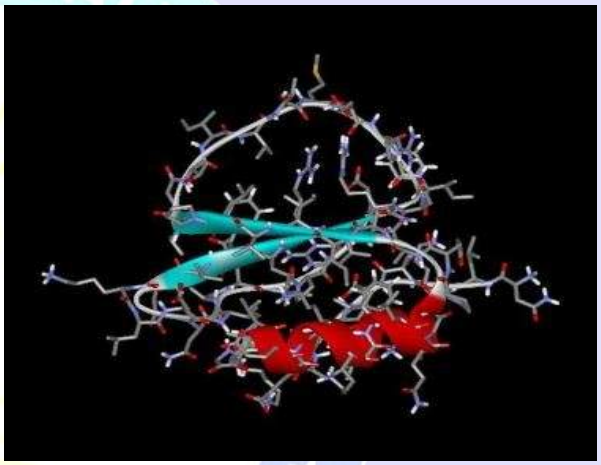


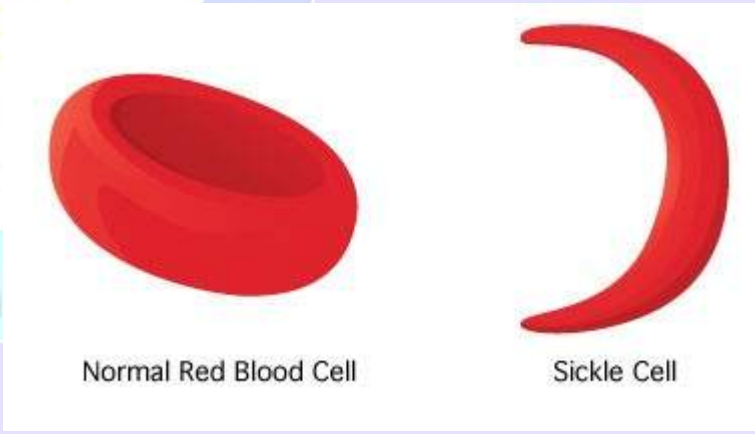
Why Each Protein Has A Unique Structure?

Amino acid sequence

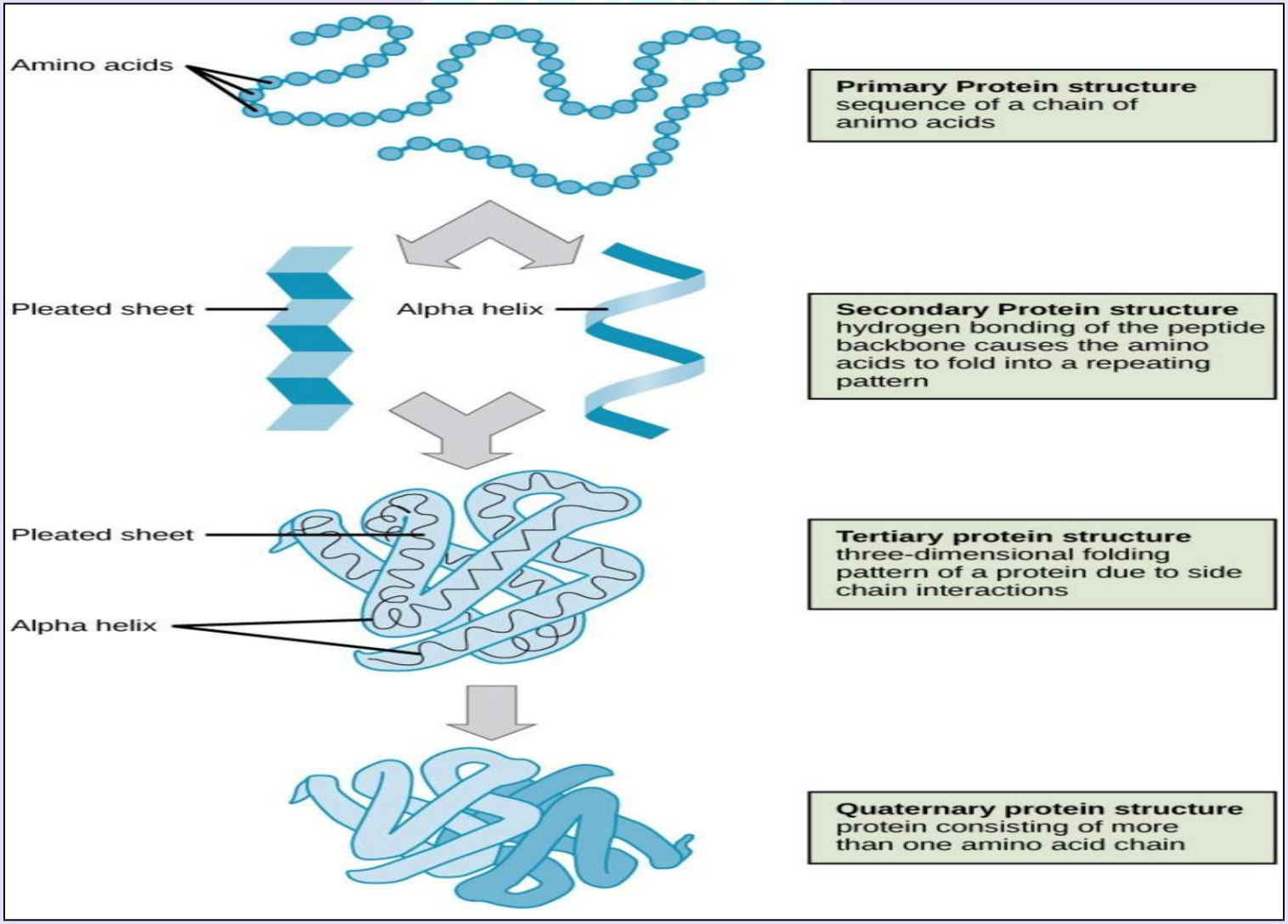
NLKTEWPELVGKSV
EEAKKVILQDKPEAQ
IIVLPVGTIVTMEYRI
DRVRLFVDKLDNIAE
V

Folding!



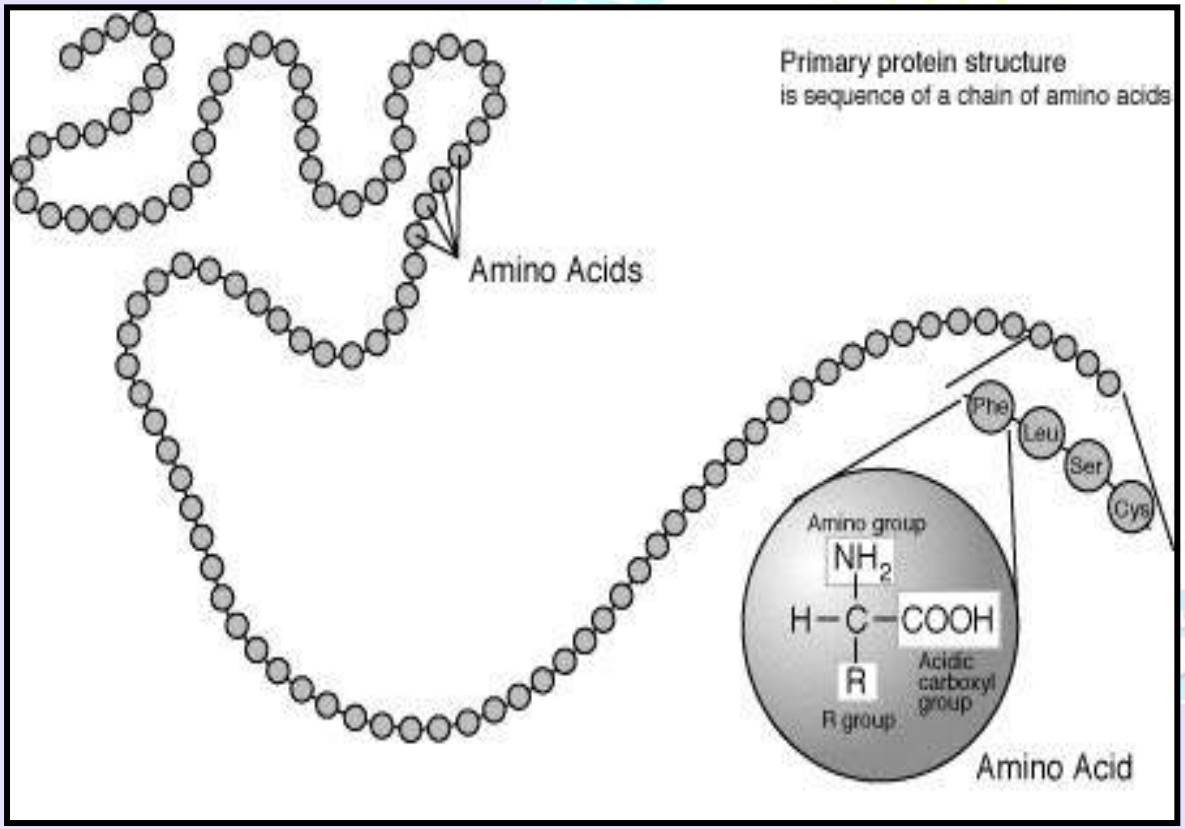


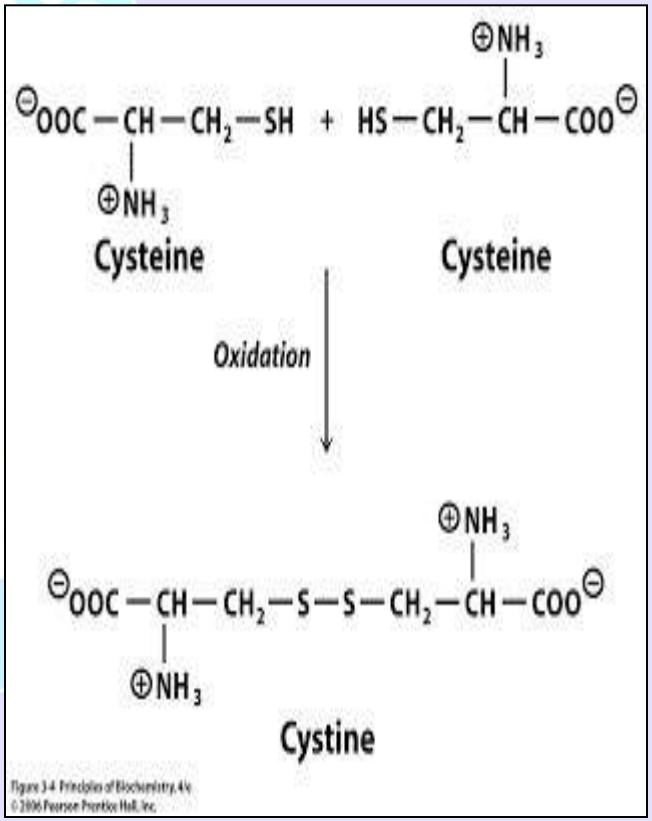
Why it is important to Investigate Protein Structure?



Protein Primary Structure

The primary structure is the sequence of amino acids that are linked together. The linear structure is called a polypeptide





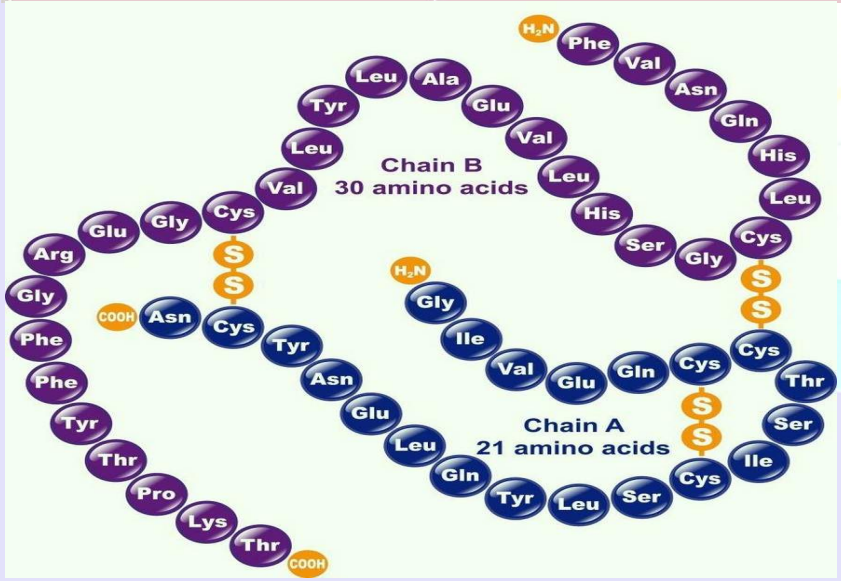
Protein Secondary Structure

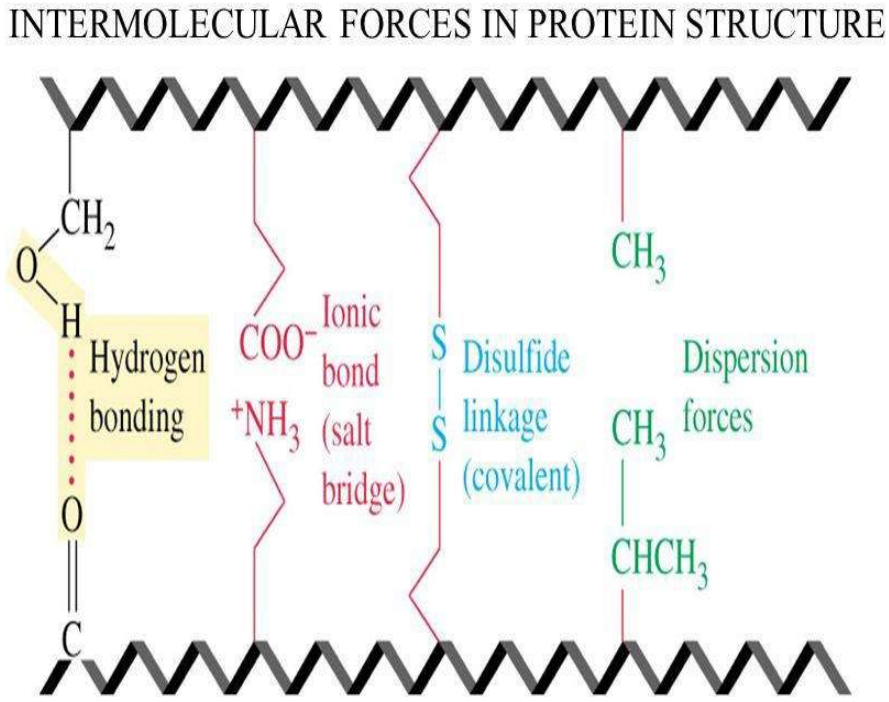
PROTEIN FOLDING

- The linear sequence of the linked amino acids contains the information necessary to generate a protein molecule with a unique three-dimensional shape
- The peptide bond allows for rotation around it and therefore the protein can fold and orient the R groups in favorable positions
- **Polar amino acids** (acidic, basic and neutral) are hydrophilic and tend to be placed on the **outside** of the protein.
- **Non-polar** ***(hydrophobic)*** amino acids tend to be placed on the **inside** of the protein

Types Of Interactions In Protein Folding

Non covalent interactions	Covalent interactions
Ionic	Peptide bonds
Hydrogen bonding	Disulphide bonds
Van der waal forces	
Hydrophobic interactions	





The 4 types of interactions that determine tertiary structure of proteins are shown. Though varying in strength, each set of interactions is significant in the formation of intrachain bonds that fixes the structure of the overal polymer.

Secondary Structures

- The folding of short (3- to 30-residue), segments of polypeptide into geometrically ordered units
- Secondary structure refers to a local spatial arrangement of the polypeptide chain
- The **α helix**
  - stabilized by hydrogen bonds between nearby residues
- The **β sheet**
  - stabilized by hydrogen bonds between adjacent segments that may not be nearby
- Irregular arrangement of the polypeptide chain is called the random coil

The α Helix

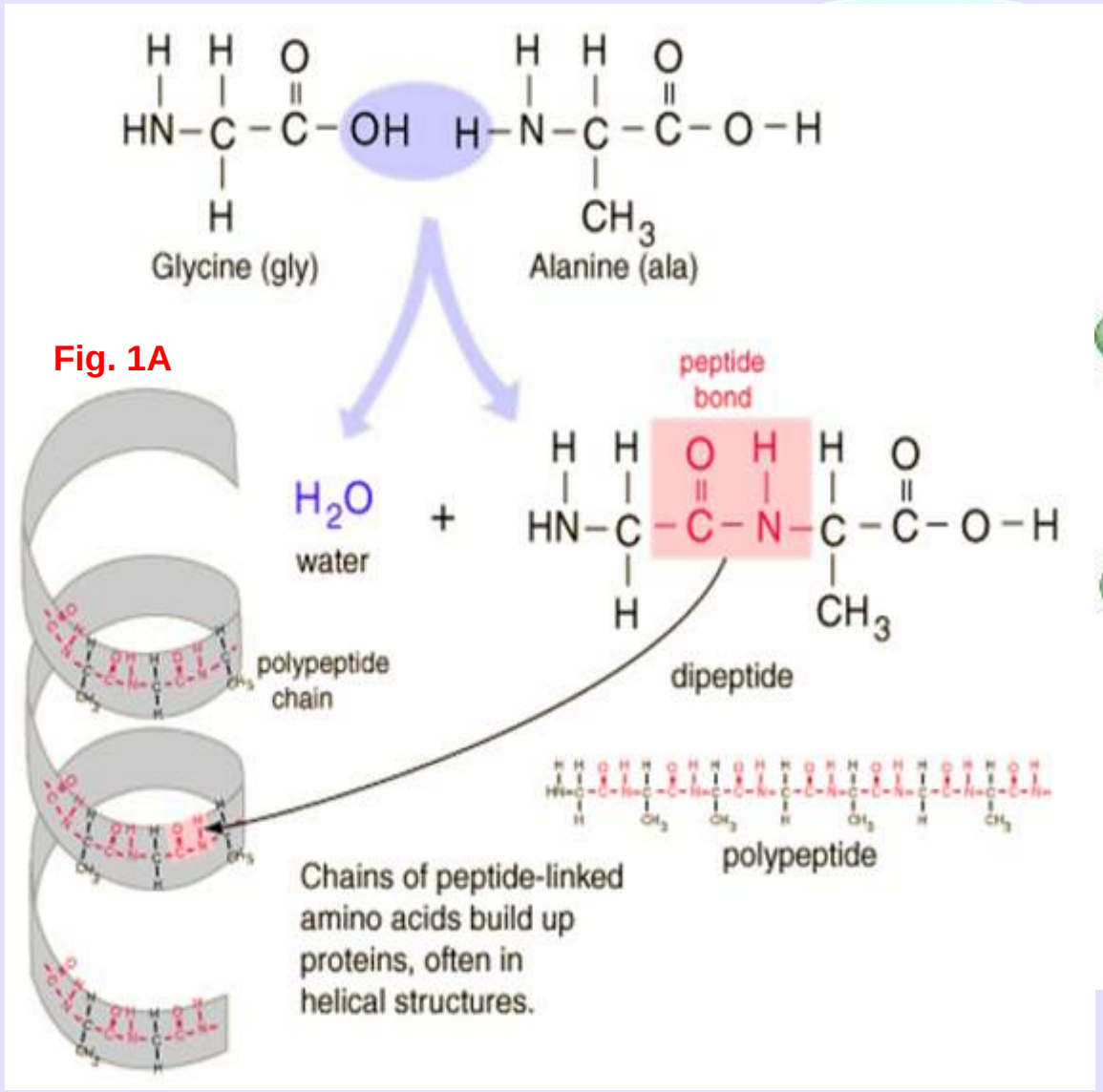
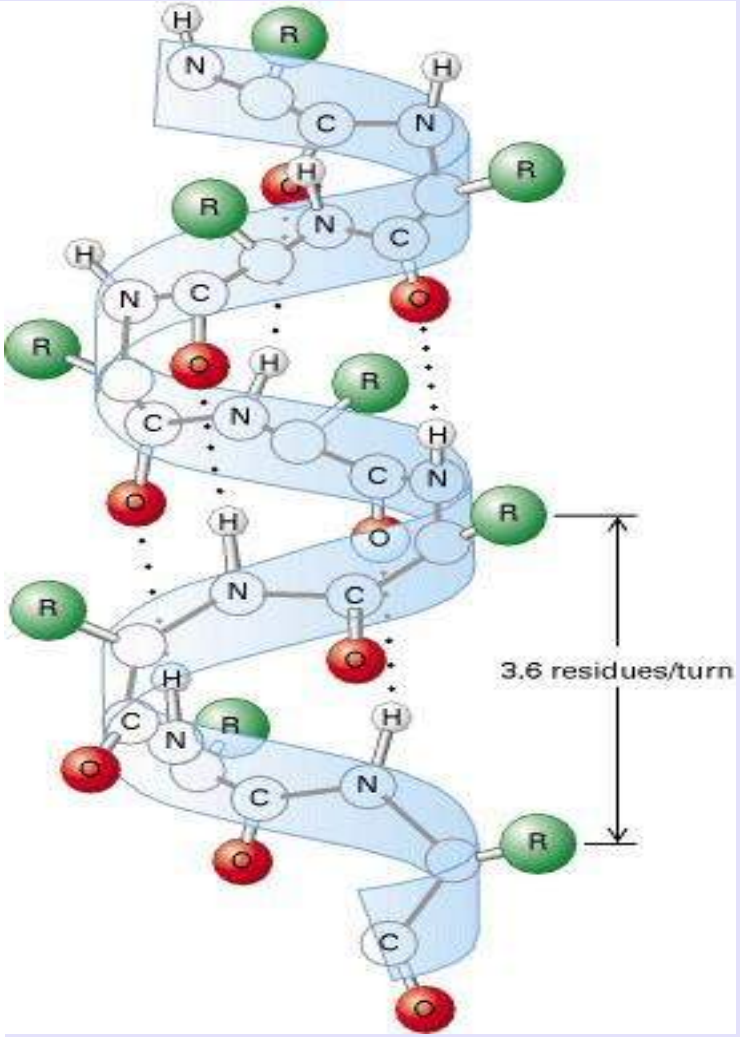


Fig. 1A

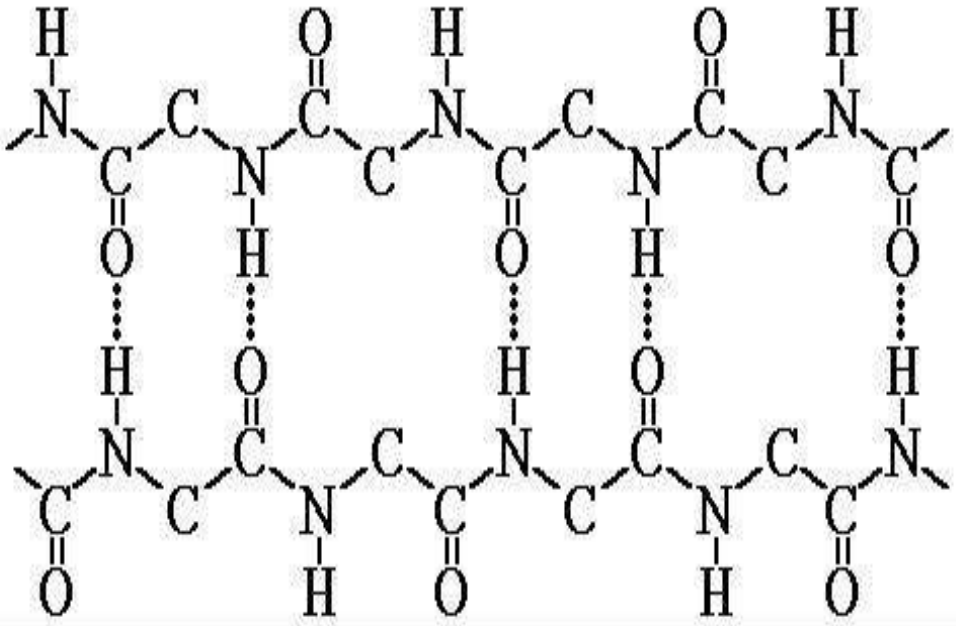


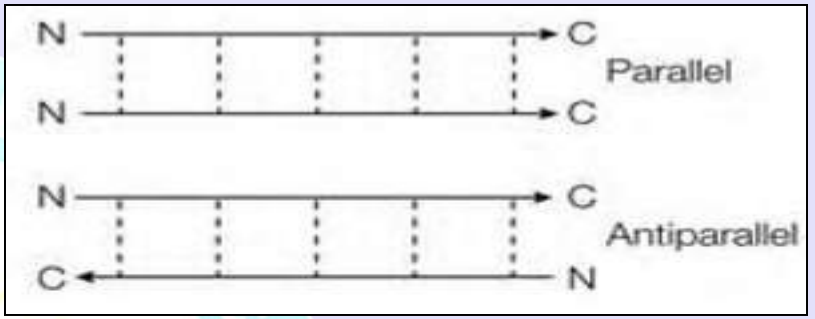
β-Sheet

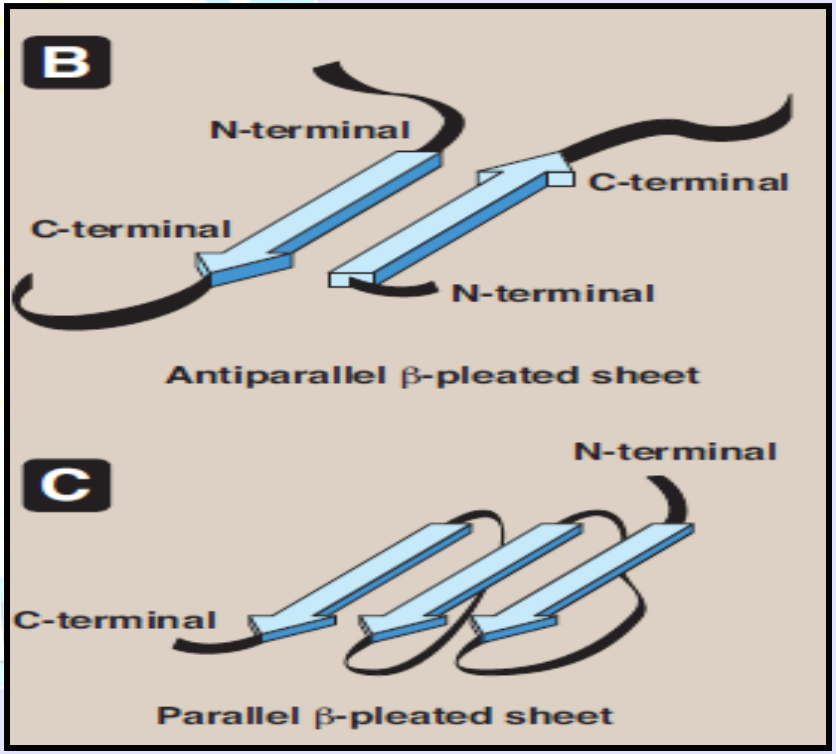
Fig. 1A

β-sheet

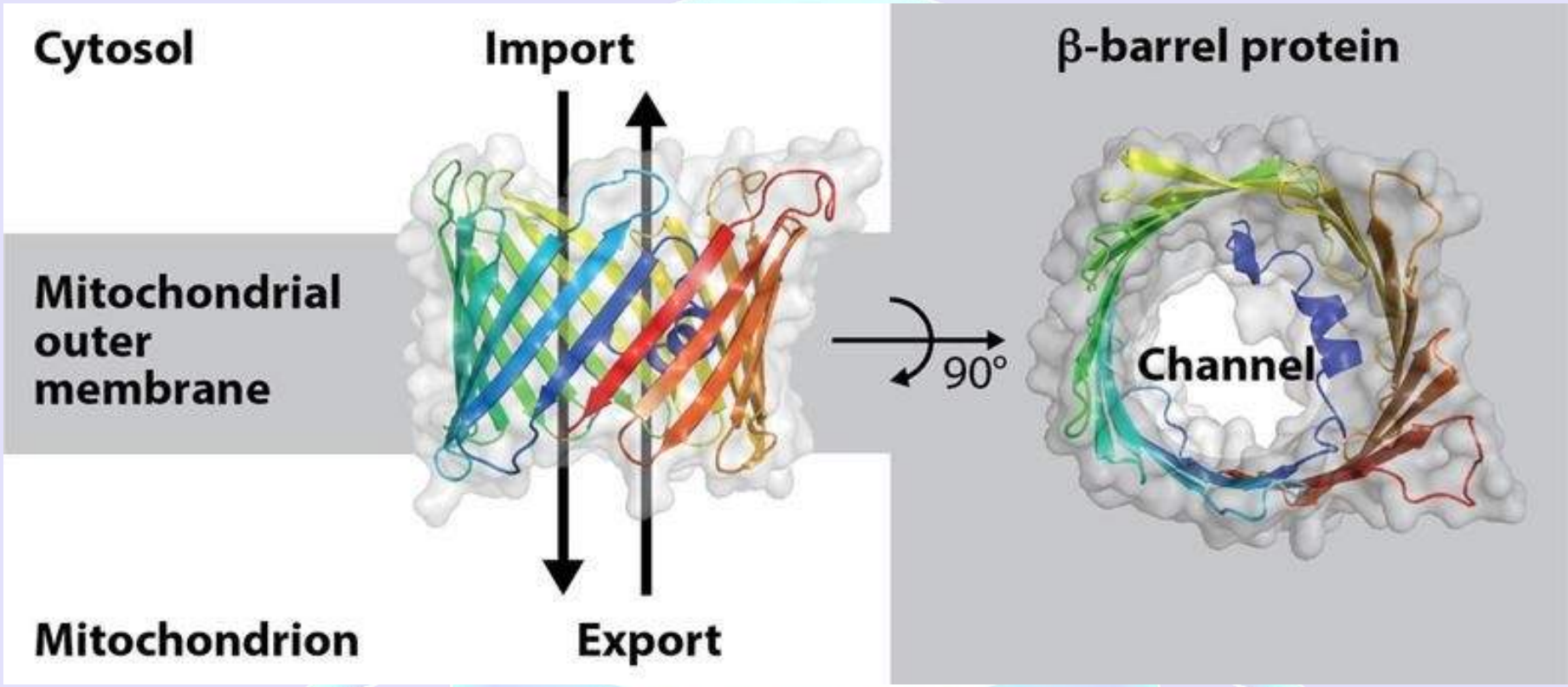
The pleated sheet structure of the β-sheet is held together by Hydrogen bonds between the amide groups of linear polypeptide chains. The average number of amino acid residues in a typical β-sheet is six with an average of six strands bonding together.



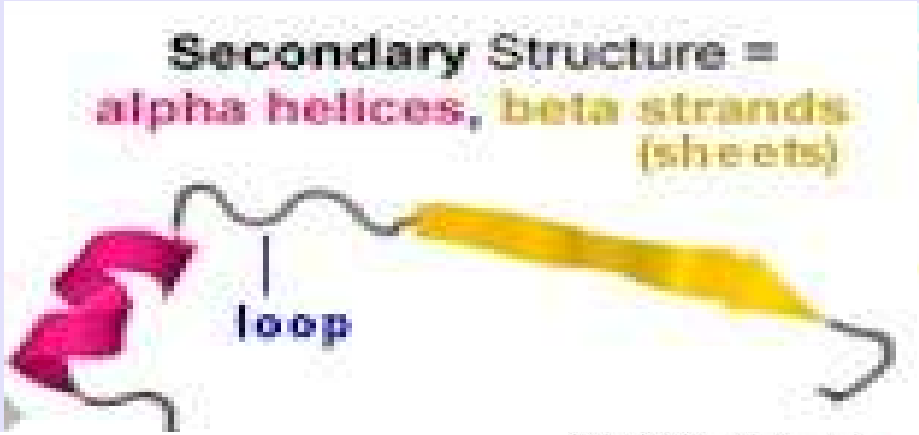


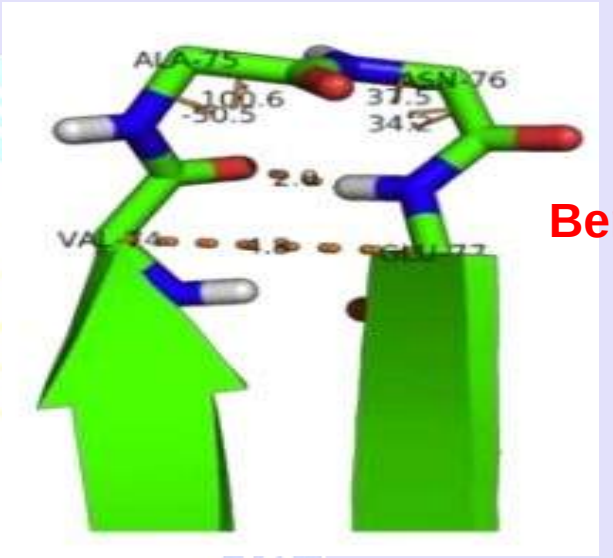


β-barrel

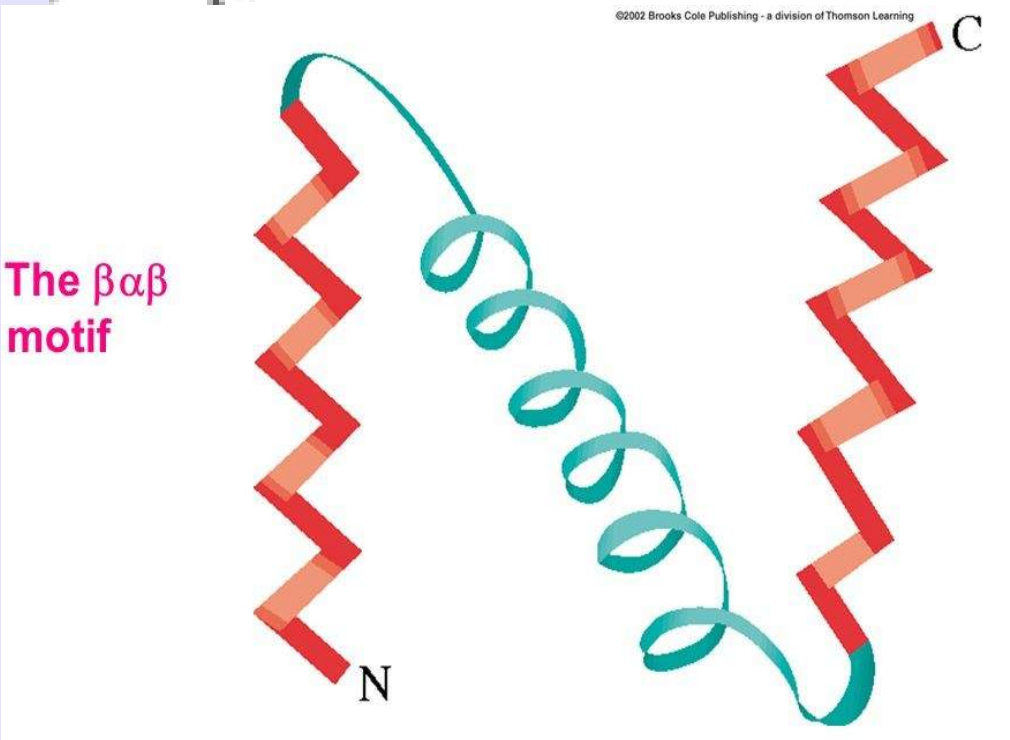

Cytosol
Import
β-barrel protein
Mitochondrial outer membrane
90°
Channel
Mitochondrion
Export

Loops, Bends, Motif and Domains



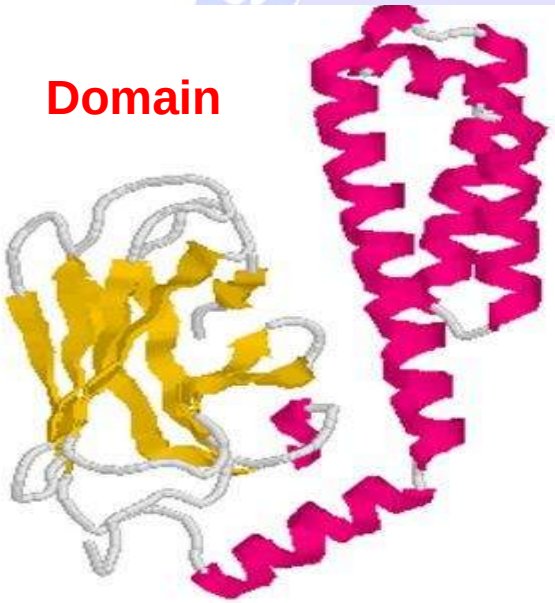


Bends



The βαβ motif

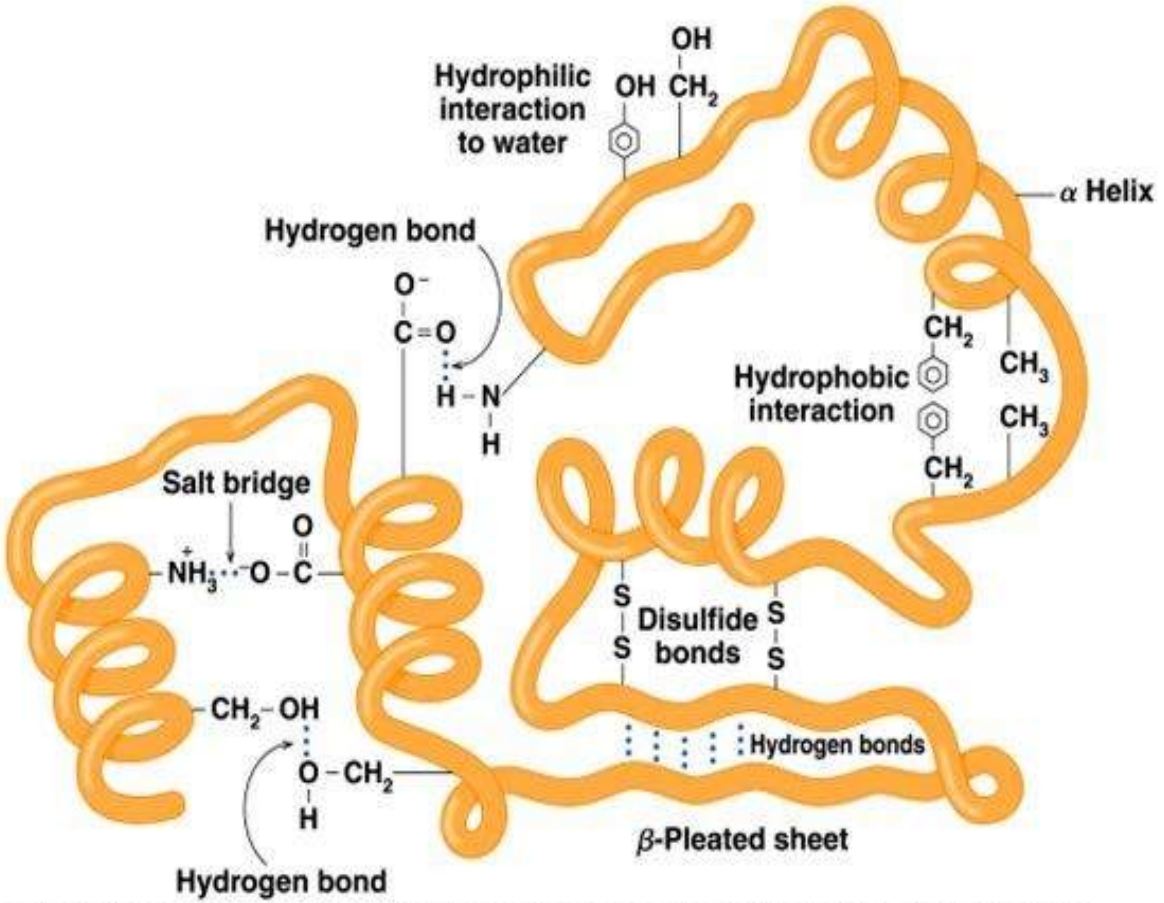
Domain



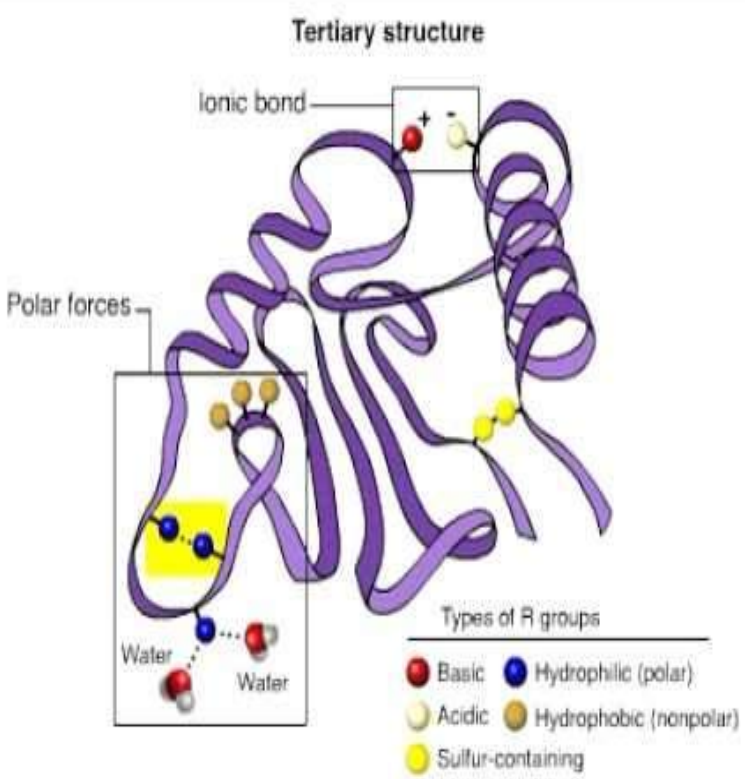
Protein Tertiary Structure

Tertiary Structure

The interactions of the R groups give a protein its specific three-dimensional tertiary structure.



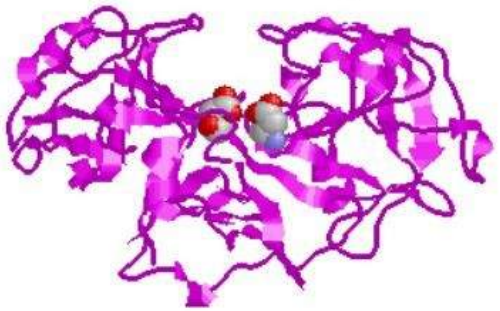
Timberlake, *General, Organic, and Biological Chemistry*. Copyright © Pearson Education Inc., publishing as Benjamin Cummings



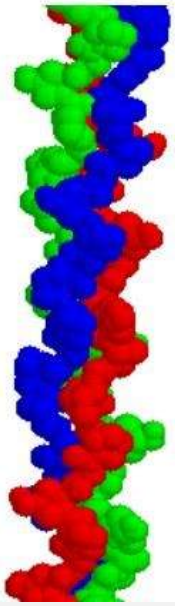
Overall 3-D shape of a protein. Two basic types are *globular* and *fibrous*.

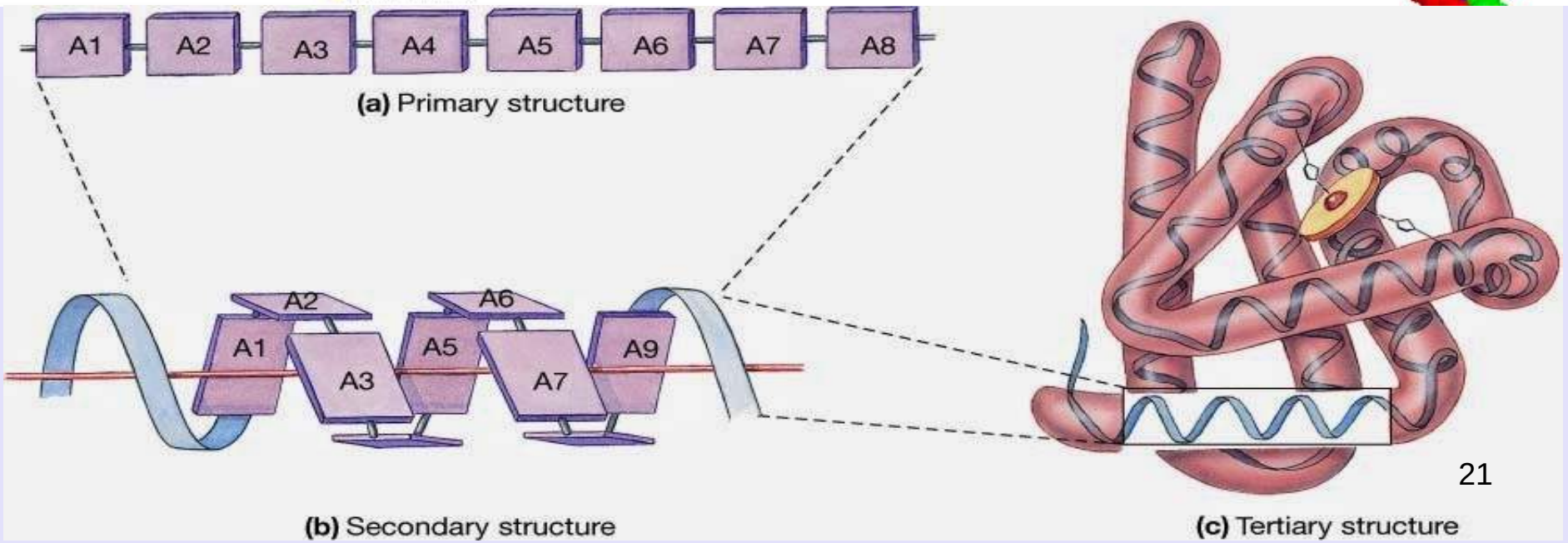
Examples:

Globular (Pepsin)



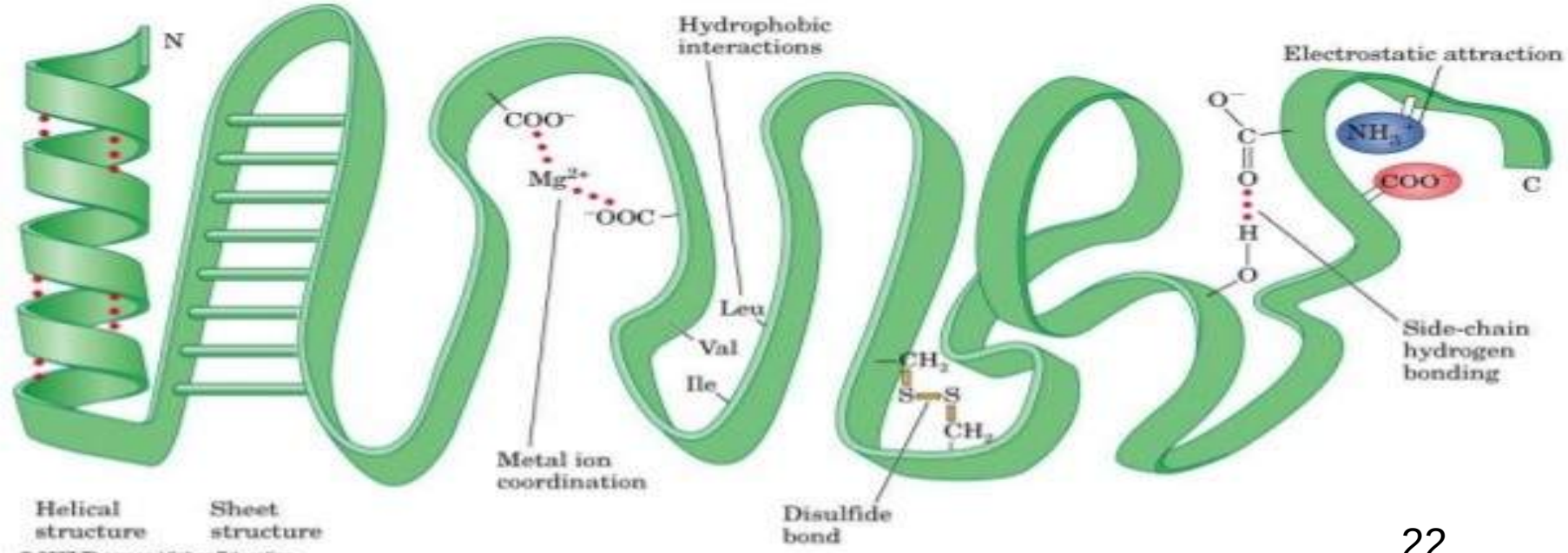
Fibrous (Collagen)





Stabilizing Forces in Tertiary Structure

Disulfide bonds	• Stabilize the native conformation of the protein.
Hydrophobic interactions	• In the interior of the protein to hide the hydrophobic residues.
Hydrogen bonds	• Formed between polar residues and water and between other polar groups within the protein.
Ionic interactions.	• To stabilize the protein structure.



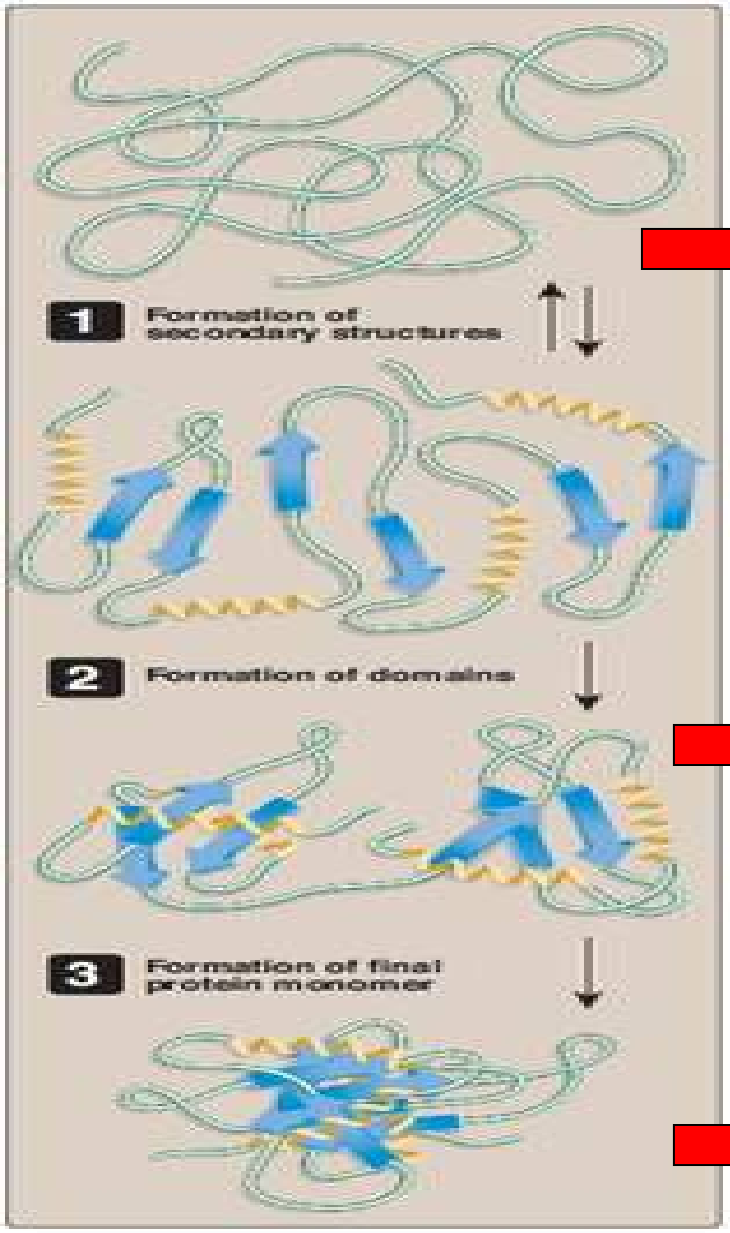


Figure 2.12
Steps in protein folding.

Forming Secondary Structur

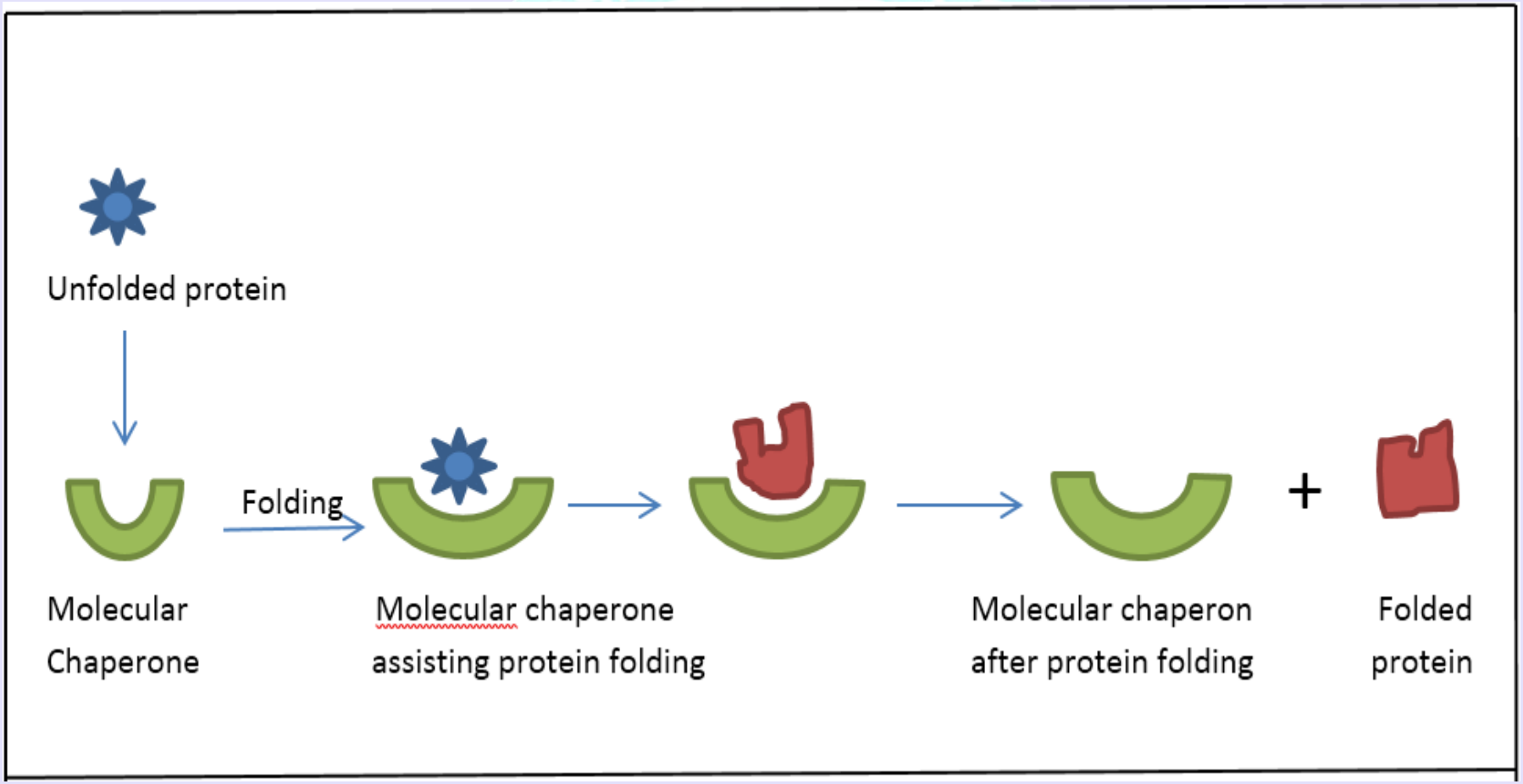
Forming DOMAINS

Forming Final Protein

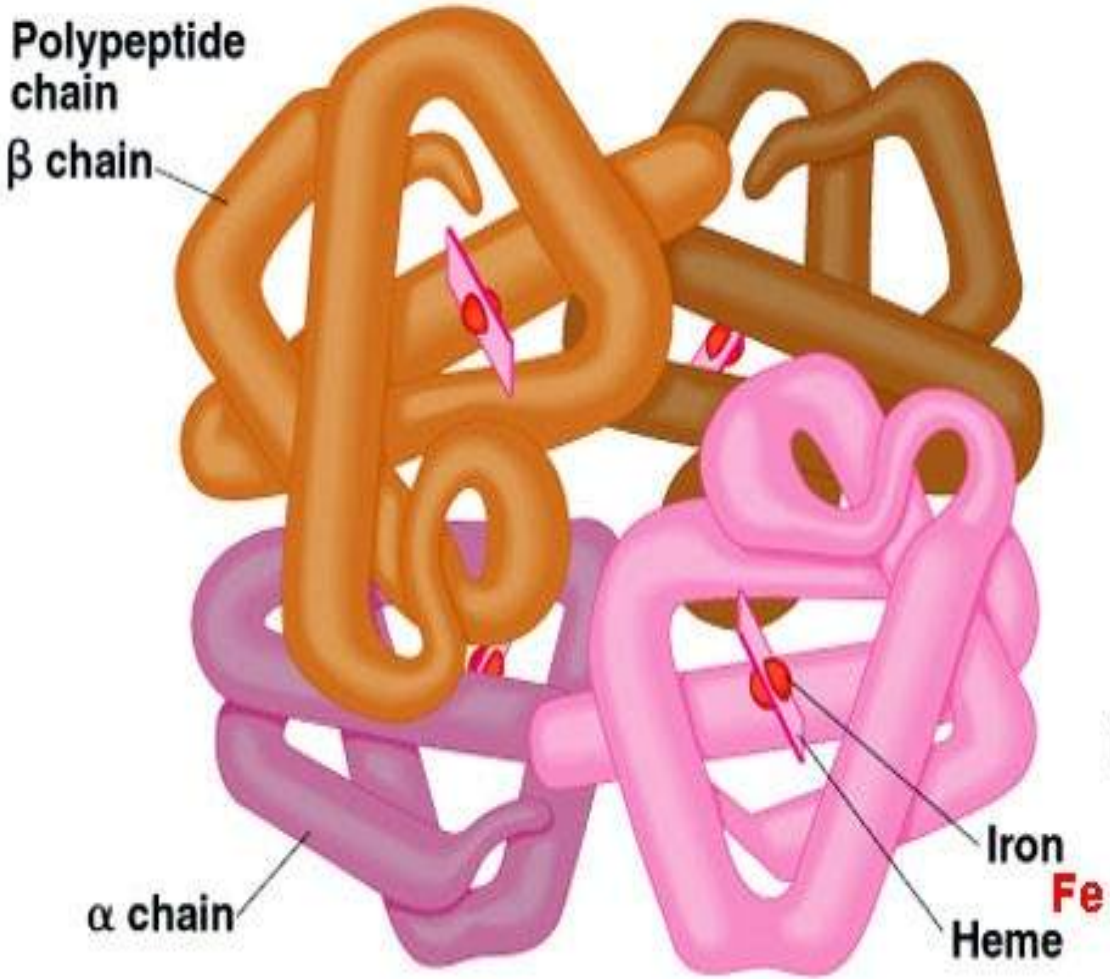
Protein Denaturation

- Protein denaturation results in the unfolding and disorganization of the protein's secondary and tertiary structures, which are **not** accompanied by hydrolysis of peptide bonds.
- Denaturing agents include heat, organic solvents, mechanical mixing, strong acids or bases, detergents, and ions of heavy metals such as lead and mercury.

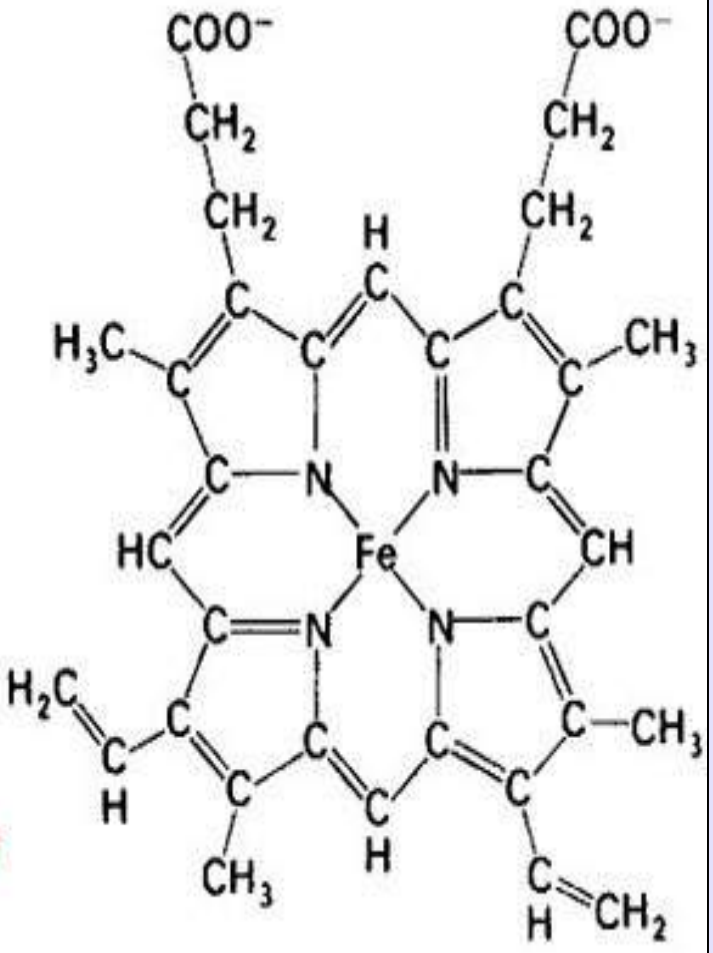
Chaperone and Folding



Protein Quaternary Structure



Hemoglobin



Heme
(Fe-protoporphyrin IX)

Protein Misfolding

1. Amyloid Disease
2. Prion Disease



Thank You